#1

$$\dot{R} = [\omega_x] R$$

$$\therefore$$

a) $$\dot{R} = \begin{bmatrix} -.866 & -.5 & 0 \\ .5 & -.866 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b) $$\dot{R} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -.5 & .866 & 0 \end{bmatrix}$$

c) $$\dot{R} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ .866 & .5 & 0 \end{bmatrix}$$

d) $$\dot{R} = \begin{bmatrix} -.474 & .218 & .371 \\ .643 & -.178 & -.631 \\ .267 & -.883 & .619 \end{bmatrix}$$

#2

$${}^0d_{01} = \begin{bmatrix} 20 \\ 15 \\ 10 \end{bmatrix} m \qquad {}^0R_1 = \begin{bmatrix} .5 & -.866 & 0 \\ .866 & .5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^0v_{01} = \begin{bmatrix} 10 \\ 5 \\ .25 \end{bmatrix} m/s \qquad {}^0a_{01} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} m/s^2$$

$${}^0\omega_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} rad/s \qquad {}^1r_{imu} = \begin{bmatrix} .1 \\ 0 \\ 0 \end{bmatrix}$$

f: ${}^1v_{01}$, ${}^1a_{01}$, ${}^1\omega_{01}$

$$v_P = v_{com} + \omega \times r$$

$$a_P = a_{com} + \cancel{\alpha}^{0} \times r + \omega \times (\omega \times r)$$

$${}^0r_{imu} = {}^0R_1 \, {}^1r_{imu} = \begin{bmatrix} .05 \\ .0866 \\ 0 \end{bmatrix} m$$

$${}^0v_{imu} = {}^0v_{01} + {}^0\omega_{01} \times {}^0r = \begin{bmatrix} 9.8008 \\ 5.115 \\ .25 \end{bmatrix}$$

$${}^1v_{imu} = {}^0R_1^T \cdot {}^0v_{imu} = \begin{bmatrix} 9.33 \\ -5.93 \\ .25 \end{bmatrix} m/s$$

$${}^0a_{imu} = {}^0a_{com} + [{}^0\omega]_x ([{}^0\omega]_x \, {}^0r) = \begin{bmatrix} -1.26 \\ -.458 \\ 1 \end{bmatrix}$$

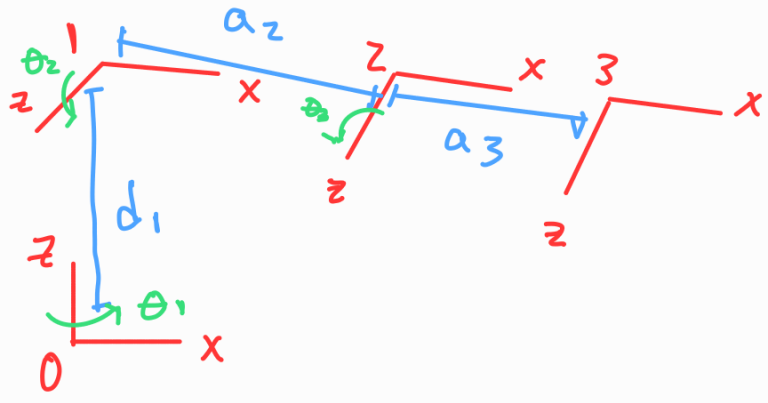
$${}^1a_{meas} = {}^0R_1^T ({}^0a_{imu} - {}^0g) = \begin{bmatrix} -1.029 \\ 0.866 \\ 10.81 \end{bmatrix} m/s^2$$

$${}^1\omega_{01} = {}^0R_1^T \, {}^0\omega_{01} = \begin{bmatrix} 0 \\ 0 \\ 2.3 \end{bmatrix} rad/s$$

$$\theta = \begin{bmatrix} \pi/3 & \pi/6 & \pi/6 \end{bmatrix}^T \text{ rad}$$

$$\dot{\theta} = \begin{bmatrix} .5 & -1.0 & 1.5 \end{bmatrix}^T \text{ rad/s}$$

L	a	d	α	θ
1	0	1	$\pi/2$	~~4~~
2	.5	0	0	~~5~~
3	.3	0	0	~~6~~



F: $\omega_i, \dot{d}_{0i}, \ddot{d}_{0i}$; $i = \{1,2,3\}$, $J_v \dot{\theta}$

$$A_i = \begin{bmatrix} C\theta & -S\theta C\alpha & S\theta C\alpha & a \cdot C\theta \\ S\theta & C\theta S\alpha & -C\theta S\alpha & a \cdot S\theta \\ 0 & S\alpha & C\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_i = \prod_i A_i$$

${}^0T_1 =$ At this point I decided to code the rest